# ISO4730: 2017 Standard

The chemical composition of tea tree oil is defined by international standard ISO 4730: 2017 which specifies levels of 15 of the more than 113 components found in pure Australian tea tree oil as well as a number of physical parameters.

Any batch of oil sold by an ATTIA member must be accompanied by an independently tested certificate of analysis demonstrating conformance to these Standards. ATTIA also recommends that the country of origin is clearly declared when purchasing tea tree oil from any supplier.

A summary of the standard:

### ISO 4730: 2017 Essential oil of *Melaleuca*, terpinen-4-ol type (Tea Tree oil)

Essential oil obtained by steam distillation of the leaves and terminal branchlets of *Melaleuca alternifolia* (Maiden et Betche) Cheel or of *M. linariifolia* Sm.

ISO 4730: 2017 also specified the Enantiomeric Distribution for TTO:

**The enantiomeric distribution for terpinen-4-ol is (R)(+) 67 % - 71 % and (S)(-) 29 % - 33 %.**

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Some essential oil components can exist in two enantiomeric forms designated as (R) or (S), D or L or (+) or (-) isomers. Many enantiomers have distinctly different properties and hence their presence in the right form is critical. Also, pure natural essential oils contain enantiomers in characteristic ratios. This ratio is upset by the addition of adulterants including synthetic major components of different enantiomeric ratios. Consequently, the measurement of enantiomeric excess or enantiomeric ratio as per ISO 22972 in an informative annex of appropriate isolates in International Standards, provides an extra measure of essential oil authenticity.

| Characteristics Requirements | | Method |
|---|---|---|
| Appearance | Clear, mobile liquid | - |
| Colour | Colourless to pale yellow | - |
| Odour | Characteristic | - |
| Relative density (20° C) | Min: 0.885 Max: 0.906 | ISO 279 |
| Refractive index (20° C) | Min: 1.475 Max: 1.482 | ISO 280 |
| Optical rotation (20° C) | between + 7° and + 12° | ISO 592 |
| Miscibility in ethanol (volume fraction)at 20° C | It shall not be necessary to use more than 2 volumes of ethanol, 85 % (volume fraction), to obtain a clear solution with 1 volume of essential oil | |
| Flashpoint (closed cup) | mean value +59° | ISO/TR 11018 |
| Min volume of test sample | 50 ml | ISO 201 |

**Chromatographic profile:**

| Component | Min % | Max % |
|---|---|---|
| α-Pinene | 1.00 | 4.00 |
| Sabinene | traces | 3.50 |
| α-Terpinene | 6.00 | 12.00 |
| Limonene | 0.50 | 1.50 |
| p-Cymene | 0.50 | 8.00 |
| 1,8-Cineole | traces | 10.00 |
| γ-Terpinene | 14.00 | 28.00 |
| Terpinolene | 1.50 | 5.00 |
| Terpinen-4-ol | 35.00 | 48.00 |
| α-Terpineol | 2.00 | 5.00 |
| Aromadendrene | 0.20 | 3.00 |
| Ledene (syn. viridiflorene) | 0.10 | 3.00 |
| δ-Cadinene | 0.20 | 3.00 |
| Globulol | traces | 1.00 |
| Viridiflorol | traces | 1.00 |

Traces: <0.01%

A full copy of this standard is available for purchase **HERE**

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